

Landslide Detection System using mbed NXP LPC 1768

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Abstract: Landslides are one of the most frequent natural disasters that occur after a heavy rainfall season. By using a landslide detection system, users can be alerted at an earlier time on when a landslide might occur. The SW-420 Vibration Sensor Module is a sensor that detects the vibrations of the surrounding. The sensor is programmed with two conditions where it would light up an LED depending on whether a vibration is detected or not. In this project, our results have shown that when a vibration is detected, a HIGH input is received by the sensor which causes the red LED to light up. When a vibration is not detected, a LOW input is received which causes the green LED to light up. By using the SW-420 Vibration Sensor Module, a simple landslide detection system can be built.

Index Terms—About; high level, low level, landslide, vibration,

I. INTRODUCTION

Landslides have become a common natural disaster in our country. A landslide occurs when the ground collapses down a slope due to gravity or being pulled by the weight of heavy rocks and soil. They usually occur in mountainous or hillside areas. One of the reasons why landslides occur is due to heavy rainfalls which happened throughout the year especially during the monsoon seasons. This has caused the soil to become too soft due to constantly being hit by rain which can lead to a landslide. In the latest news, the landslide that occurred near a campsite in Batang Kali had claimed around 31 lives where the majority of them consisted of young children. The cause of death in landslides is mainly due to people not being informed earlier on when a landslide would occur nearby. By alerting people on when a landslide would happen, we will be able to reduce the number of deaths from a landslide. This can be done by using a landslide detection system.

A landslide detection system is a device that is capable of detecting the vibrations of the surroundings using a vibration sensor. The main purpose of a landslide detection system is to monitor the frequency of vibration of an area and to inform the user when a landslide would occur. In this project, a SW-420 Vibration Sensor Module is used as the vibration sensor. The SW-420 Vibration Sensor Module works by detecting a vibration and provides either digital value which is either HIGH or LOW depending on the input received. For this

assignment, the SW-420 was programmed with the following conditions. When a vibration is detected, the Red LED will be switched on. When a vibration is not detected, the Green LED will be switched on instead.

II. MBED NXP LPC 1768 DEVICE

The mbed Microcontrollers are a family of development boards for ARM microcontrollers intended for quick prototyping. The mbed NXP LPC1768 Microcontroller is designed for prototyping a wide variety of devices, particularly those with Ethernet, USB, and the flexibility of a large number of peripheral ports and FLASH memory. It is packaged in a compact DIP form factor for prototyping on through-hole PCBs, stripboard, and breadboard, and has an integrated USB FLASH programmer.

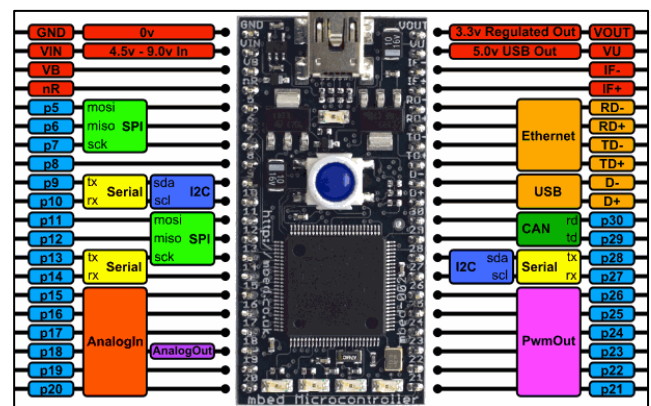


Figure 1: mbed NXP LPC 1768 pinout

The NXP LPC1768, on which it is based, features a 32-bit ARM Cortex-M3 core operating at 96MHz. It has 512KB FLASH, 32KB RAM, and numerous interfaces, including Ethernet, USB Host and Device, CAN, SPI, I2C, ADC, DAC, PWM, and more I/O interfaces. The above pinout displays the common interfaces and their locations. Note that all numbered pins (p5-p30) are also 'Digital In' and 'Digital Out' interfaces.

The mbed Microcontrollers give seasoned embedded developers a robust and productive environment for creating proofs-of-concept. For developers new to 32-bit microcontrollers, mbed offers an approachable prototype

option with the backing of common libraries, tools, and support from the mbed community.

III. METHODOLOGY

A. Block Diagram

The block diagram of the system is shown in Figure 2. When the vibration sensor has reached a stable state, the circuit will be activated, and the output will be at a high level. The circuit will momentarily become disconnected, and the output will be set to a low state whenever the movement or vibration takes place. Once the vibration in the soil is detected, the microcontroller will then decide to send a signal to the respective LED. The condition of LEDs is based on two conditions. The experiment overviewed two cases:

1. When vibration is **NOT DETECTED**, **LED green is ON**.
2. When measured vibration is **DETECTED**, **LED red is ON**

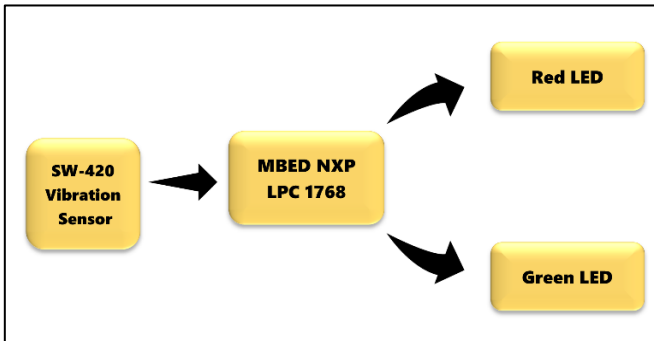


Figure 2:Block Diagram of Landslide Detection System

B. Project Design

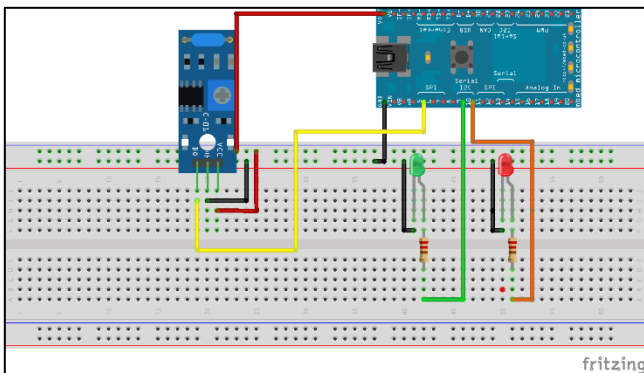


Figure 3: Breadboard Circuit designed in Fritzing

Figure 3 shows the complete circuit diagram of the SW420 (vibration sensor) with the LPC1768 microcontroller. The SW-420 is a non-directional vibration sensor with great sensitivity. The VCC and GND pin of the vibration sensor was connected to the power and ground in the breadboard respectively. The DO (digital output) pin was connected to pin 5 of the LPC1768 microcontroller. The Green led was

connected to pin 9 and the red led was connected to pin 10 of the microcontroller. Red led indicates a critical situation, whereas green led signifies a stable one. When everything is normal, a green led will turn on, and if there is any vibration, a red led will turn on. The circuit is activated and the output is high when the module is stable. When the movement or vibration occurs, the circuit will be briefly disconnected and output low.

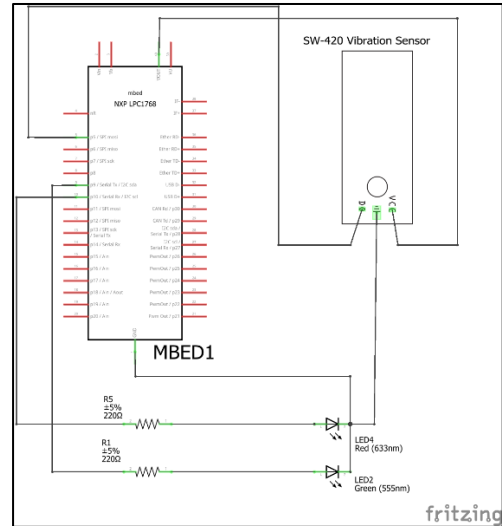


Figure 4: Schematic Circuit designed in Fritzing

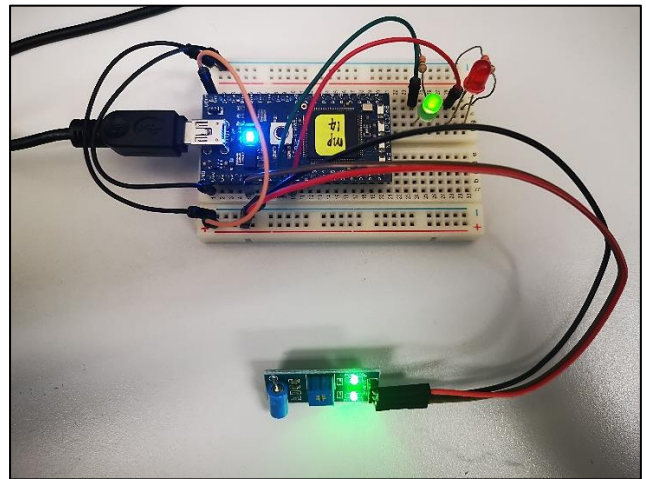


Figure 5: Hardware Construction of the Landslide Detection System

The hardware design comprises the circuit components depicted in Figure 5. This system's architecture can be considered economical, as it uses only four components: an SW-420 vibration sensor, mbed NXP LPC 1768, resistors, and LEDs.

IV. SOFTWARE STRUCTURE

A. Flowchart

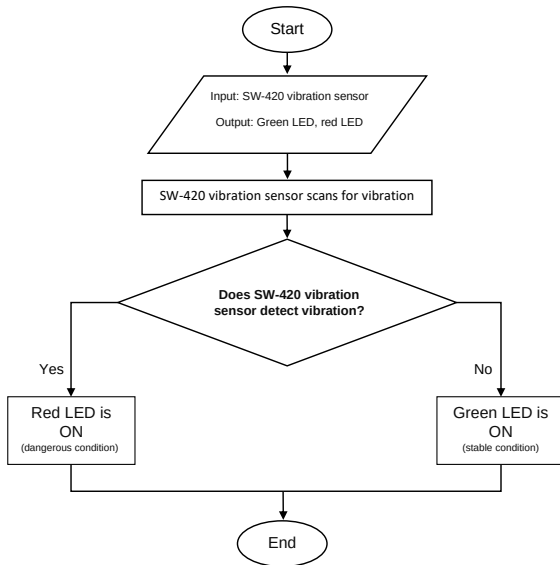


Figure 6: Flowchart of the Landslide Detection System

The whole working flow of the project appears in the above flowchart. The input of the circuit is the SW420 vibration sensor and the output are green and red LEDs. If SW420 detects any vibrations it will turn on the red led which indicates the dangerous condition occurs. If not, it just turns on the green led which indicates a stable condition.

B. Project Code

```

test15.s*
1  AREA MYCODE, CODE, READONLY ; code area (segment)
2  ENTRY ; start of area MYCODE
3  EXPORT SystemInit ; export SystemInit symbol
4  SystemInit ; SystemInit routine
5
6  EXPORT __main ; export __main symbol
7  __main
8
9  ;setup pin (refer PINSEL0 table)
10 PINSEL0 EQU 0x4002C000 ; PINSEL0 address
11 LDR R0, =PINSEL0 ; move PINSEL0 address into R0
12 LDR R1, =0x00000000 ; set all PINSEL0 pins as GPIO
13 STR R1, [R0]
14
15 ;set pin direction: input(0), output(1)
16 FIODIR EQU 0x2009C000 ; FIODIR address
17 LDR R0, =FIODIR
18 LDR R1, =0x0000000F ; output: (P9,P10); input:P5
19 STR R1, [R0]
20
21 ;set pin to be affected: affected(0), unaffected(1)
22 FIOOMASK EQU 0x2009C010 ; FIOOMASK address
23 LDR R0, =FIOOMASK
24 LDR R1, =0xFFFFFFFF ; affected: p5,p9,p10
25 LDR R1, =0xFFFFFFFF ; affected: p5,p9,p10
26 STR R1, [R0]
27
28
29 ;===== read sensor at p5 =====
30
31 check ;check sensor
32 FIOOPIN EQU 0x2009C014 ; FIOOPIN address
33 LDR R0, =FIOOPIN ; move address into R0
34 LDR R1, [R0] ; move data in memory address R0 (FIOOPIN) into R1
35 AND R1, R1, #0x00000200 ; replace data in FIODIR with pin P5 address
36 CMP R1, #0 ; if input pin P5 == 0,
37 BEQ stable ; branch to stable
38
39 ;turn off pin: no change(0), clear(1)
40 FIOOCLR EQU 0x2009C01C
41 LDR R0, =FIOOCLR
42 LDR R1, =0x00000001 ; P9 GREEN (OFF)
43 STR R1, [R0]
44
45 ;turn on pin: no change(0), set(1)
46 FIOOSET EQU 0x2009C018
47 LDR R0, =FIOOSET
48 LDR R1, =0x00000002 ; P10 RED (ON)
49 STR R1, [R0]
50 B check ;branch to
51
52 stable
53 ;FIOOCLR EQU 0x2009C01C
54 LDR R0, =FIOOCLR
55 LDR R1, =0x00000002 ; P10 RED (OFF)
56 STR R1, [R0]
57
58 ;FIOOSET EQU 0x2009C018
59 LDR R0, =FIOOSET
60 LDR R1, =0x00000001 ; P9 GREEN (ON)
61 STR R1, [R0]
62 B check
63
64 ALIGN
65 END
  
```

V. PROJECT DEMONSTRATION/RESULTS

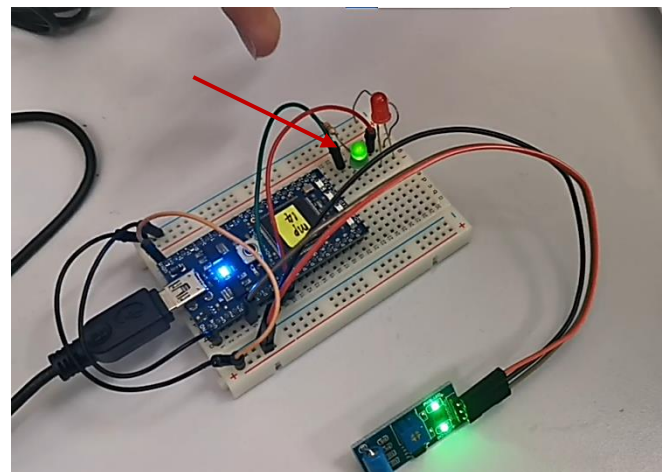


Figure 7: When there is no vibration or shake is detected, the Green LED is ON and Red LED is at OFF state

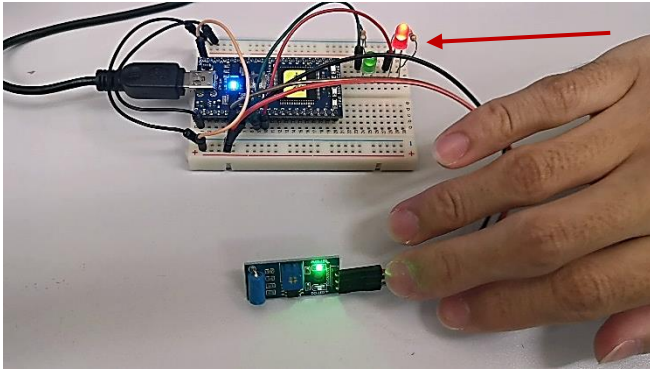


Figure 8: When there is no vibration or shake is detected, the Red LED is ON and Green LED is at OFF state

VI. CONCLUSION

This study provided a microcontroller-based architecture for a simple landslide detection system. The system offers help to authorities to save people in the landslide-prone area as soon as signs of vibration are detected. The system's monetary design ensures the lowest possible production costs. The simulation findings indicate that the technology operates as intended in conditions. To improve this system, additional sensors can be added to monitor landslide occurrence based on geotechnical instrumentations.

VII. REFERENCES

1. mbed LPC1768 | Mbed. (n.d.). <https://os.mbed.com/platforms/mbed-LPC1768/>
2. Zuo, B. (n.d.). Grove - Vibration Sensor(SW-420) - Seeed Wiki. https://wiki.seeedstudio.com/Grove-Vibration_Sensor_SW-420/
3. Indian researchers working on early landslide detection systems to reduce fatalities, damage. (n.d.). The Economic Times. Retrieved January 21, 2023, from <https://economictimes.indiatimes.com/news/india/indian-researchers-working-on-early-landslide-detection-systems-to-reduce-fatalities-damage/articleshow/87236375.cms>
4. Chenhui Wang, Wei Guo, Kai Yang, Xi Wang and QingJia Meng, 6 June 2022, "Real-Time Monitoring System of Landslide Based on LoRa Architecture" <https://www.frontiersin.org/articles/10.3389/feart.2022.899509/full>
5. Malaysia Now, 27 December 2022, "Batang Kali landslide Malaysia's second worst disaster in terms of death" <https://www.malaysianow.com/news/2022/12/27/batang-kali-landslide-malysias-second-worst-disaster-in-terms-of-deaths>
6. The Guardian, 17 December 2022, "Malaysia landslide kills at least 21 campers as recovery efforts continue" <https://www.theguardian.com/world/2022/dec/17/malaysia-landslide-kills-at-least-21-campers-as-recovery-efforts-continue>
7. Masato Ohki, Takahiro Abe, Takeo Tadono and Masanobu Shimada, 13 May 2020, "Landslide detection in mountainous forest areas using polarimetry and interferometric coherence" <https://earth-planet-space.springeropen.com/articles/10.1186/s40623-020-01191-5>
8. Sourav Gupta, 23 April 2019, "Interfacing Vibration Sensor Module with Arduino" <https://circuitdigest.com/microcontroller-projects/arduino-sw-420-vibration-sensor-module-interfacing>
9. Ikhwan El Akmal Pakpahan, Poltak Sihombing and Mahyuddin K. M. Nasution, 2020, "Analysis of the SW-420 Vibration Sensor Performance on Vibration Tools by using a Fuzzy Logic Method" <https://www.scitepress.org/Papers/2020/103360/103360.pdf>